

Linux Lab 5

By Jai Pandey (260840140011)

1. Using the top command

```
top - 14:36:57 up 3:14, 1 user, load average: 0.21, 0.34, 0.27
Tasks: 261 total, 1 running, 260 sleeping, 0 stopped, 0 zombie
%Cpu(s): 1.3 us, 0.2 sy, 0.0 ni, 97.7 id, 0.0 wa, 0.5 hi, 0.2 si, 0.0 st
MiB Mem : 7648.9 total, 5597.9 free, 1311.8 used, 1022.9 buff/cache
MiB Swap: 4096.0 total, 4096.0 free, 0.0 used. 6337.1 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
2587	csf	20	0	5194124	395396	139144	S	2.0	5.0	2:36.88	gnome-shell
2844	csf	20	0	389148	40352	33060	S	1.3	0.5	3:39.16	vmtoolsd
750	root	20	0	458200	9608	7820	S	1.0	0.1	3:25.73	vmtoolsd
733	root	20	0	82824	4564	4096	S	0.7	0.1	0:04.11	irqbalance
34311	csf	20	0	225936	4348	3452	R	0.7	0.1	0:00.32	top
2783	csf	20	0	981348	113492	50596	S	0.3	1.4	0:18.85	gnome-soft+
4072	root	20	0	0	0	0	I	0.3	0.0	0:08.56	kworker/u5+
4336	root	20	0	0	0	0	I	0.3	0.0	0:16.20	kworker/2:+
34225	csf	20	0	769668	55900	43984	S	0.3	0.7	0:01.05	gnome-term+
1	root	20	0	174860	17248	11172	S	0.0	0.2	0:17.35	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.07	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workq+
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-+
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-+
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-+
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-+

2. Using the ps aux command

```
[csf@localhost ~]$ ps aux
```

USER	PID	%CPU	%MEM	VSZ	RSS	TTY	STAT	START	TIME	COMMAND
root	1	0.1	0.2	174860	17248	?	Ss	11:22	0:17	/usr/lib/systemd
root	2	0.0	0.0	0	0	?	S	11:22	0:00	[kthreadd]
root	3	0.0	0.0	0	0	?	S	11:22	0:00	[pool_workqueue_
root	4	0.0	0.0	0	0	?	I<	11:22	0:00	[kworker/R-rcu_g
root	5	0.0	0.0	0	0	?	I<	11:22	0:00	[kworker/R-sync_
root	6	0.0	0.0	0	0	?	I<	11:22	0:00	[kworker/R-slub_
root	7	0.0	0.0	0	0	?	I<	11:22	0:00	[kworker/R-netns
root	9	0.0	0.0	0	0	?	I<	11:22	0:00	[kworker/0:0H-ev
root	10	0.0	0.0	0	0	?	I	11:22	0:00	[kworker/u512:0-
root	11	0.0	0.0	0	0	?	I<	11:22	0:00	[kworker/R-mm_pe
root	12	0.0	0.0	0	0	?	I	11:22	0:00	[kworker/u512:1-
root	13	0.0	0.0	0	0	?	I	11:22	0:00	[rcu_tasks_kthre
root	14	0.0	0.0	0	0	?	I	11:22	0:00	[rcu_tasks_rude_
root	15	0.0	0.0	0	0	?	I	11:22	0:00	[rcu_tasks_trace
root	16	0.0	0.0	0	0	?	S	11:22	0:00	[ksoftirqd/0]
root	17	0.0	0.0	0	0	?	I	11:22	0:01	[rcu_preempt]
root	18	0.0	0.0	0	0	?	S	11:22	0:00	[rcu_exp_par_gp_
root	19	0.0	0.0	0	0	?	S	11:22	0:00	[rcu_exp_gp_kthr
root	20	0.0	0.0	0	0	?	S	11:22	0:00	[migration/0]
root	21	0.0	0.0	0	0	?	S	11:22	0:00	[idle_inject/0]
root	23	0.0	0.0	0	0	?	S	11:22	0:00	[cpuhp/0]

3. Using pidof <program>

```
[csf@localhost ~]$ pidof firefox
34759 34725 34669 34651 34619 34598 34566 34563 34515 34455 34385
[csf@localhost ~]$
```

4. Using kill -9 <PID>

```
[csf@localhost ~]$ pidof firefox
34759 34725 34669 34651 34619 34598 34566 34563 34515 34455 34385
[csf@localhost ~]$ kill -9 34759
[csf@localhost ~]$ pidof firefox
34725 34669 34651 34619 34598 34566 34563 34515 34455 34385
[csf@localhost ~]$
```

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3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workq+
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-+
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-+
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-+
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-+

6. SSH Configuration

a. Enabled SSH Daemon to auto-start on StartUp and checked its status

```
[root@localhost ~]# systemctl enable sshd
[root@localhost ~]# systemctl status sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enable)
   Active: active (running) since Tue 2026-09-01 16:50:18 IST; 22h ago
     Docs: man:sshd(8)
           man:sshd_config(5)
  Main PID: 964 (sshd)
    Tasks: 1 (limit: 48546)
   Memory: 2.0M (peak: 2.2M)
      CPU: 11ms
   CGroup: /system.slice/sshd.service
           └─964 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 01 16:50:18 localhost.localdomain systemd[1]: Starting OpenSSH server daemon...
Sep 01 16:50:18 localhost.localdomain sshd[964]: Server listening on 0.0.0.0 port 22.
Sep 01 16:50:18 localhost.localdomain sshd[964]: Server listening on :: port 22.
Sep 01 16:50:18 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
lines 1-16/16 (END)
```

b. Disabled root login

```
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes
```

c. Created a SSH key of type ed25519

```
[root@localhost ~]# ssh-keygen -t ed25519
Generating public/private ed25519 key pair.
Enter file in which to save the key (/root/.ssh/id_ed25519):
Enter passphrase for "/root/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_ed25519
Your public key has been saved in /root/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:pXQ2uRSDIeLJR5jkQA0hNm0i/mnNH13XTJ0Kl0Ujnt4 root@localhost.localdomain
The key's randomart image is:
+--[ED25519 256]--+
|o+++.ooo .oo . o |
|. *o+ .. +. + o |
|.o = . . B o = |
|o . . . * oo 0 . |
|+.      S .o * E |
|=      . . o |
|. o . . |
| ...o . |
|.o .. |
+-----[SHA256]-----+
[root@localhost ~]#
```

d. Verified SSH Daemon configuration

```
[root@localhost ~]# sshd -t
[root@localhost ~]#
```

e. Systemctl restart & status

```
[root@localhost ~]# systemctl restart sshd
[root@localhost ~]# systemctl status sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-09-02 16:24:33 IST; 40s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
  Main PID: 36206 (sshd)
    Tasks: 1 (limit: 48546)
   Memory: 1.3M (peak: 1.5M)
      CPU: 108ms
   CGroup: /system.slice/sshd.service
           └─36206 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 02 16:24:33 localhost.localdomain systemd[1]: Starting OpenSSH server daemon...
Sep 02 16:24:33 localhost.localdomain sshd[36206]: Server listening on 0.0.0.0 port 22.
Sep 02 16:24:33 localhost.localdomain sshd[36206]: Server listening on :: port 22.
Sep 02 16:24:33 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
[root@localhost ~]#
```

f. Viewing past login logs via /var/log/secure

```
cationAgent, locale en_IN.UTF-8) (disconnected from bus)
Sep  1 16:51:42 localhost su[3345]: pam_unix(su:session): session opened for user root(uid=0) by csf(uid=1000)
Sep  2 11:31:07 localhost su[3345]: pam_unix(su:session): session closed for user root
Sep  2 11:33:22 localhost su[3804]: pam_unix(su:session): session opened for user root(uid=0) by csf(uid=1000)
Sep  2 11:57:57 localhost gdm-password][3905]: gkr-pam: unlocked login keyring
Sep  2 12:03:45 localhost gdm-password][3989]: gkr-pam: unlocked login keyring
"/var/log/secure" 44L, 5243B
```